

Paul Edwin Curtis

210 Whitney Ave.,
New Haven, Connecticut 06511.
paul.curtis@yale.edu (+1) 203 909-8932

EDUCATION	YALE UNIVERSITY	2021 - 2026
	Doctor of Philosophy (Ph.D., MPhil.), Atmosphere, Ocean, Climate Dynamics. (<i>Advanced to candidacy May 2023; Ph.D. expected Fall 2026</i>). Primary Academic Advisor: Professor Alexey V. Fedorov. Thesis: The Response of the Global Ocean Meridional Overturning Circulation to Radiative Forcing: Regime Shifts, Oscillations, and Equilibrium States.	
	IMPERIAL COLLEGE LONDON	2017 - 2021
	Master of Science (MSci.), Physics.	
SELECTED FELLOWSHIPS	NASA FUTURE INVESTIGATORS (FINESST) GRADUATE FELLOWSHIP	2023 - 2026
	<i>Competitive federal fellowship supporting research in Earth and space science</i>	
	GEOPHYSICAL FLUID DYNAMICS FELLOWSHIP, WHOI	2024
	<i>Highly-selective interdisciplinary program in theoretical geophysical fluid dynamics</i>	
	WORLD CLIMATE RESEARCH PROGRAMME (WCRP) FELLOWSHIP, YALE JACKSON SCHOOL OF GLOBAL AFFAIRS, YALE UNIVERSITY	2025 - 2026
	<i>Inaugural fellowship supporting interdisciplinary climate science collaboration</i>	
SELECTED AWARDS	PHILIP M. ORVILLE PRIZE, YALE UNIVERSITY	2026
	<i>In Recognition of "Outstanding Research and Scholarship in the Earth Sciences"</i>	
	ELIAS LOOMIS PRIZE, YALE UNIVERSITY	2024
	<i>Awarded for "Excellence in Studies of Physics of the Earth"</i>	
	LUDLAM PRIZE, IMPERIAL COLLEGE LONDON	2021
	<i>Awarded for "Excellence in Studies of Atmospheric Physics"</i>	
PEER-REVIEWED PUBLICATIONS	4. P. E. Curtis , & A. V. Fedorov (2025), Dansgaard-Oeschger Events Under Global Cooling Without Continental Ice Sheets. <i>Journal of Climate</i> , 38 (17), 4299–4313.	
	3. P. E. Curtis , & A. V. Fedorov (2024b), Collapse and Slow Recovery of the Atlantic Meridional Overturning Circulation (AMOC) Under Abrupt Greenhouse Gas Forcing. <i>Climate Dynamics</i> , 62 (7), 5949–5970.	
	2. P. E. Curtis , & A. V. Fedorov (2024a), Spontaneous Activation of the Pacific Meridional Overturning Circulation (PMOC) in Long-Term Ocean Response to Greenhouse Forcing. <i>Journal of Climate</i> , 37 (5), 1551-1565.	
	1. P. E. Curtis , P. Ceppi, & G. Zappa (2020), Role of the Mean State for the Southern Hemispheric Jet Stream Response to CO ₂ Forcing in CMIP6 Models. <i>Environmental Research Letters</i> , 15 (6), 064011.	
PUBLICATIONS UNDER REVIEW	P. E. Curtis , A. V. Fedorov, & N. Feldl, Constant Equilibrium Climate Sensitivity (ECS) in Ultralong Simulations of a Wide Range of Climates in CESM1. Manuscript in revision for <i>Climate Dynamics</i> .	
	P. E. Curtis , T. Schiminovich [†] , A. V. Fedorov, & W. Liu, Millennial Variability and Abrupt Changes with Origins in the Southern Ocean. Manuscript in revision. ([†] <i>undergraduate advisee</i>).	

PUBLICATIONS IN PREPARATION	P. E. Curtis, M.-L. Timmermans, A. J. K. Yang, & S. T. Cole, Partitioning the Influences of Surface-layer and Intrahalocline Processes on Episodic Mechanically-driven Mixing at the Base of the Arctic Ocean’s Surface Mixed Layer. To be submitted to <i>Geophysical Research Letters</i>.	
	P. E. Curtis, A. V. Fedorov, Playing the Long Game: Deep Ocean Equilibration Timescales Quantified through Ideal Age Tracer. To be submitted to <i>Geophysical Research Letters</i>.	
	P. E. Curtis, A. V. Fedorov, How Does Labrador Sea Deep Convection Influence Weakening of the Atlantic Meridional Overturning Circulation (AMOC) in a Warming Climate? To be submitted to <i>Geophysical Research Letters</i>.	
NON-REFEREED PUBLICATIONS	Paul Curtis, “Idealised Models of Moist Convection”, in <i>WHOI GFD Summer Program Annual Proceedings</i>, 2024.	
	Paul Curtis & Cy David, “Lecture Notes for ‘Weakly Nonlinear Theory for Convective Instabilities’”, in <i>WHOI GFD Summer Program Annual Proceedings</i>, 2024.	
SELECTED RESEARCH EXPERIENCE	PH.D. MAJOR DISCOURSE	2021 - Present
	<i>Department of Earth and Planetary Sciences, Yale University</i>	
	• Led an investigation into transient and equilibrium responses of the Global Ocean Meridional Overturning Circulation (GMOC) to greenhouse gas forcing. • Independently conducted a suite of numerical simulations using a full-complexity Earth system model, exceeding field-leading integration lengths by up to $\sim 4\times$. • Identified several previously-unseen abrupt climate regime shifts and oscillations under different boundary conditions, inaccessible from shorter simulations. • Skills: independent climate model integration (CESM1); HPC environments; large-scale ($>100\text{TB}$) data analysis; matlab; python.	
	PH.D. MINOR DISCOURSE	2021 - Present
	<i>Department of Earth and Planetary Sciences, Yale University</i>	
	• Investigated event-driven turbulent heat fluxes in the upper Arctic Ocean using observational oceanographic data from autonomous instruments. • Developed robust analysis frameworks to process irregular and noisy field data using advanced filtering techniques to isolate event-driven dynamics. • Identified shallow intrahalocline eddies as a dominant driver of elevated heat fluxes impacting seasonal ice growth, and performed a novel computation for the vertical diffusivity of ocean heat in the Arctic. • Skills: observational oceanographic data (ITP; BGOS); matlab; python.	
	GEOPHYSICAL FLUID DYNAMICS (GFD) FELLOWSHIP	2024
	<i>Woods Hole Oceanographic Institution (WHOI)</i>	
	• Initiated a seven-week collaborative research project around the program’s central theme: ‘multiscale geophysical fluid dynamics’ to construct low-order analytical models for atmospheric convection. • Employed advanced multiscale asymptotic analysis and pseudospectral methods to form a new extension to reduced models of rotating convection incorporating the effects of fluid heating by condensation, reproducing several physical aspects of atmospheric convection. • Skills: numerical modeling (Dedalus); multiscale asymptotic analysis; python.	

	UNDERGRADUATE RESEARCH FELLOWSHIP (UROP) <i>Department of Physics, Imperial College London</i>	2019
	• Ten-week project analysing projected 21st-century Southern Hemisphere atmospheric circulation changes in large ensembles of IPCC-class Earth system models. • Identified up to a 90% reduction in circulation bias in CMIP6 relative to CMIP5 and a relationship between model resolution and projected circulation change. • Skills: climate model analysis (CMIP5; CMIP6; ERA5; CAM4); python; R.	
CLIMATE POLICY EXPERIENCE	WORLD CLIMATE RESEARCH PROGRAMME GRADUATE FELLOWSHIP <i>Yale Jackson School of Global Affairs, Yale University</i>	2025 - Present
	• Co-organiser of the new bi-semester ‘Science-policy Interchange’ seminar series, connecting students and researchers across climate science and public policy. • Regularly publish science communication articles on meetings. • Generating unique engagements with leaders in global environmental governance. • Initiated a research project investigating the impacts of tropical forest land-use changes on ocean circulation and climate using CESM2.	
SELECTED ADDITIONAL TRAINING & WORKSHOPS	• 30th ANNUAL COMMUNITY EARTH SYSTEM MODEL (CESM) WORKSHOP <i>National Center for Atmospheric Research (NCAR), Boulder, Colorado</i> • 4th SUMMER SCHOOL ON THEORY, MECHANISMS AND HIERARCHICAL MODELING OF CLIMATE DYNAMICS <i>International Centre for Theoretical Physics (ICTP), Trieste, Italy</i> • ATLANTIC OVERTURNING: NEW OBSERVATIONS AND CHALLENGES <i>The Royal Society, London, UK</i> • UNIQ+ GRADUATE ACCESS PROGRAMME <i>University of Oxford, UK (moved online)</i>	2025 2023 2022 2020
TEACHING & MENTORSHIP	ACADEMIC SUPERVISION (WITH PROFESSOR ALEXEY V. FEDOROV) • Aaron Parr (Physics), undergraduate researcher (Dean’s Fellow). • Theo Schiminovich (EPS), senior thesis (awarded departmental prize). GUEST LECTURER • Earth System Science for Public Policy, Yale Jackson School GLBL7165. TEACHING FELLOWSHIPS • Physical Oceanography, Yale College EPS335. (<i>voluntary position</i>) • Physical Oceanography, Yale College EPS335. • Dynamic Earth, Yale College EPS110.	2026 - Present 2024 - 2025 Nov 2025 Fall 2023 Fall 2022 Fall 2021
SELECTED CONFERENCE ENGAGEMENTS	PRESENTED TALKS • 30th Annual CESM Workshop, Boulder, CO. • AGU Fall Meeting, Washington, D.C. • AGU Ocean Sciences Meeting, New Orleans, LA. • Nansen Legacy Symposium, Tromsø, Norway. • EGU General Assembly, Vienna, Austria. (<i>online</i>) • AGU Fall Meeting, Chicago, IL. OTHER RESEARCH TALKS • EPS Graduate Student Seminar, Yale University, CT. • Summer Program in GFD Final Presentation, Woods Hole, MA. • Fedorov/Jin joint group meeting, Yale University, CT.	Jun 2025 Dec 2024 Feb 2024 Nov 2023 Apr 2023 Dec 2022 Dec 2025 Aug 2024 Apr 2023

ACADEMIC
SERVICE

LEADERSHIP & ENGAGEMENT

- AOCD external committee representative, Yale EPS. 2026
- AOCD seminar organiser, Yale EPS. 2025 - Present
- IDEA committee leader, Yale EPS. 2024 - Present
- Colloquium graduate student representative, Yale EPS. 2024 - 2025
- Graduate mentoring scheme, Yale EPS. 2023 - 2025
- Faculty search student volunteer, Yale EPS. 2023 - 2025
- Disability, Mental Health, and Chronic Illness committee member, Yale EPS. 2022
- 1st generation & international student committee member, Yale EPS. 2022
- Physics undergraduate mentoring scheme, Imperial College London. 2018 - 2019

PEER-REVIEW

- Journal of Geophysical Research: Oceans.
- Ocean Science.

SKILLS

CLIMATE AND NUMERICAL MODELING

- Community Earth System Model (CESM1, CESM2).
- Dedalus.
- Coupled Model Intercomparison Project (CMIP5, CMIP6).
- Estimating the Circulation and Climate of the Ocean (ECCOv4).

OBSERVATIONAL AND REANALYSIS DATA

- Ice Tethered Profilers (ITP/WHOI).
- Beaufort Gyre Moorings (BGOS/WHOI).
- ECMWF reanalysis (ERA5).
- NASA satellite climate observation stack.

DATA ANALYSIS

- MATLAB. • Python. • R. • Command line/Linux.

SELECTED
RELEVANT
COURSEWORK

YALE UNIVERSITY

- Geophysical Fluid Dynamics (EPS 421).
- Climate Dynamics (EPS 523).
- Physical Oceanography (EPS 535).
- Energy, Mass, and Momentum Processes (ENAS 603).
- The Science of Complex Systems (EPS 528).
- Classical Statistical Thermodynamics (EPS 666).
- Applied Numerical Methods for Differential Equations (ENAS 748).

YALE UNIVERSITY (AS INVITED GUEST STUDENT)

- Co-existing with Complexity: Emerging Environmental Governance (GLBL 7155).
- Negotiating International Agreements: The Case of Climate Change (GLBL 7020).

IMPERIAL COLLEGE LONDON

- Atmospheric Physics.
- Environmental Physics.
- Advanced Hydrodynamics.
- Advanced Classical Physics.
- General Relativity.
- Quantum Field Theory.